TEACHER GUIDE · GRADE 3 · DEBUGGING

The Wrong Turn

Quick facilitation notes & answer key

Goal

Students debug turtle shape loops where the TURN angle is wrong so the shape will not close (C3.2). Correct angles are given (square 90, triangle 120, pentagon 72, hexagon 60).

Curriculum (Ontario Math 2020, Strand C3)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential, concurrent, and repeating events.

C3.2 — read and alter existing code, including code that involves sequential, concurrent, and repeating events, and describe how changes to the code affect the outcomes.

Materials

Python Turtle projected (or a web playground); a turn chart (square 90, triangle 120, pentagon 72, hexagon 60); worksheet + pencil per student.

Run it (about 30 min)

1. Run the triangle with right(90) — the open/odd shape shows the bug.
2. Change to right(120); it closes. Post the turn chart for the shapes.
3. Students do Exercises 1–4 (Ex 4 is the stretch), solo or in pairs.
4. Share answers; land the big idea: if it won't close, fix the turn.

Answer key (checked by the run-gate)

Ex 1 — right(120).

Ex 2 — the turn is wrong; change right(90) to right(60).

Ex 3 — right(72).

Ex 4 (challenge) — no, it will not close; right(60) is a hexagon turn. Use right(90) for a square.

Watch for & differentiate

Common slip: changing the range when the TURN is the bug.

Support: give the turn chart and have students match shape → turn.

Extend: ask the turn for an octagon if it does not close (45).

Success looks like

The student recognises an unclosed shape as a wrong-turn bug and fixes the angle to match the shape (Ex 1–3), explaining why a mismatched turn fails (Ex 4).